

COURIER MANAGEMENT SYSTEM(CMS)

7. PROJECT DOCUMENTATION PHASE

7.1 Introduction

Project documentation is an important phase of the Courier Management System. It records the complete information about the project, including the problem identified, requirements, design, development, testing, results, and future enhancements. Proper documentation helps the development team, users, and evaluators understand how the system was planned, developed, and tested.

The documentation provides a structured description of the Courier Management System and acts as a reference for future maintenance and improvements.

7.2 Purpose of Documentation

The main purpose of project documentation is to:

- Explain the objective and scope of the Courier Management System.
- Describe the problems faced in the existing courier process.
- Document the functional and non-functional requirements.
- Explain the system design and architecture.
- Describe the modules and features of the system.
- Record the development and testing activities.
- Present the results and outcomes of the project.
- Provide guidance for future maintenance and enhancement.

7.3 Documentation Structure

The Courier Management System documentation is organized into the following sections:

7.3.1 Project Introduction

This section introduces the Courier Management System and explains its purpose. It describes how the system helps manage customers, shipments, delivery agents, branches, tracking, invoices, notifications, and reports.

7.3.2 Problem Statement

This section explains the problems in the existing courier management process, such as:

- Manual shipment management.
- Difficulty in tracking parcels.
- Delayed delivery updates.
- Manual delivery agent assignment.
- Lack of automatic notifications.
- Difficult branch coordination.
- Manual invoice generation.
- Limited performance monitoring.

7.3.3 User Story

The user story describes the requirements from the perspective of different users.

For example:

As a customer, I want to track my shipment using a tracking ID so that I can know the status and location of my parcel.

Other users include administrators, courier staff, delivery agents, and branch managers.

7.3.4 Requirement Analysis

This section documents the requirements of the system.

Functional requirements include:

- Customer management.
- Shipment booking.
- Delivery agent assignment.
- Real-time status updates.
- Invoice generation.
- Reports and dashboards.
- Customer notifications.
- Delivery confirmation.

Non-functional requirements include:

- Security.

- Performance.
- Reliability.
- Usability.
- Scalability.
- Availability.
- Maintainability.
- Accuracy.

7.3.5 System Design

The system design explains how the Courier Management System is structured.

It includes:

- System architecture.
- Data Flow Diagram (DFD).
- Customer journey map.
- Database design.
- Module design.
- User interface design.

The design provides a clear understanding of how data moves between customers, staff, delivery agents, databases, and system processes.

7.3.6 Testing Documentation

Testing documentation records how the system was tested and whether the features worked correctly.

Testing includes:

- Unit Testing.
- Integration Testing.
- System Testing.
- User Acceptance Testing.

Test cases are documented with the **test case ID, feature, input, expected result, actual result, and status.**

Example:

Test Case	Feature	Expected Result	Status
TC01	Customer Registration	Customer should be registered successfully	Pass
TC02	Shipment Booking	Shipment should be created successfully	Pass
TC03	Tracking	Correct shipment status should be displayed	Pass
TC04	Invoice	Invoice should be generated correctly	Pass
TC05	Login	Valid user should access the system	Pass

7.3.7 Future Enhancements

The documentation also records possible improvements for future versions.

Future enhancements may include:

- Mobile application.
- GPS-based live tracking.
- AI-based delivery time prediction.
- Online payment integration.
- SMS and email notifications.
- QR-code-based shipment tracking.
- Advanced analytics.
- Cloud deployment.
- Multi-language support.

7.4 Documentation Maintenance

Project documentation should be updated whenever major changes are made to the system. New features, bug fixes, database changes, testing results, and future improvements should be recorded properly.

Maintaining updated documentation makes it easier for developers and users to understand the current version of the system.